02:13 a.m.

Proportionate Holds

Hold the victim's money inside the golden hour, and only the victim's share of it.

ExaThon India 2026 · Track 01: Money, Risk & Regulation

Team Data Dynamos

Y.V.S.S.SHUBAM

The problem

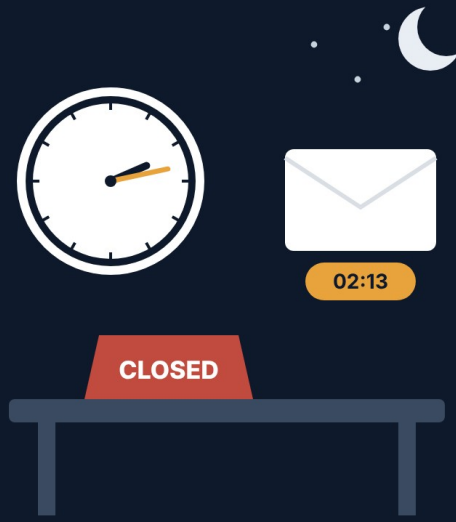
WHO



Bank nodal officer

Gets the police ticket. Must find the payment and hold what is left.

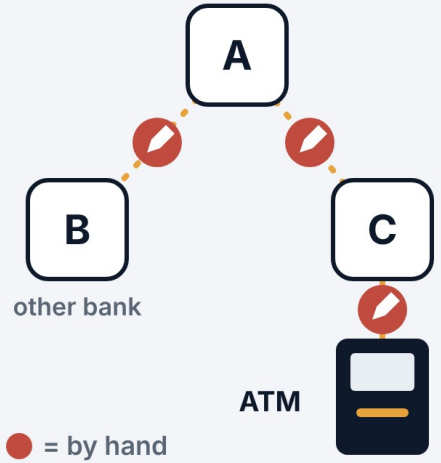
CAN'T



Waits for the desk

Night-time complaints wait for the desk to open.

BECAUSE



Hop by hop, by hand

Every hop is traced, held and typed into the portal by hand.

IF THIS WORKED



Held early and fairly

Only the victim's share, before it is withdrawn. Honest customers keep their accounts.

A bank nodal officer can't act on a cyber-fraud complaint within the golden hour, while the money is still in the first account, because night-time complaints wait for the desk to open and every hop is then traced and held by hand. If this worked, the victim's money would be held before it is withdrawn, and only the victim's share of it, so honest customers further down the trail keep their accounts.

Why it matters

1 hour

Police advise reporting within the first hour, while the money is still in the first one or two accounts.

The Week

29%

of reported money was held nationally in H1 2026.

Anantam IAS

2.18%

₹167 crore of ₹7,647 crore held (Apr 2021 to Nov 2025) had been returned to victims.

MHA SOP, 2 Jan 2026, para 2

105

petitions against account freezes settled by one High Court in a single judgment.

MediaNama

Night-time complaints miss the hold. In Rajasthan, a complaint made at night missed the hold because bank representatives work banking hours. The420 · Over-freezing is the rational choice: the notice threatens action for not complying, and nothing happens for holding too much.

The 2:13 a.m. ticket, worked the way it is today

01:52 — Lakshmi pays a fake officer ₹2,00,000 into account A at Bank X

01:55 — A splits it: ₹70,000 to B at Bank Y, ₹80,000 to C at Bank X

02:04 — C pays Sri Murugan Fuels ₹1,000 for diesel

02:13 — The police ticket reaches Bank X. The desk is closed

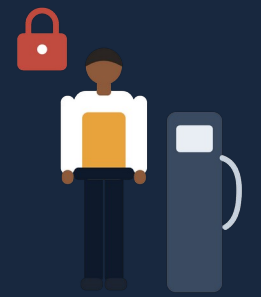
04:30 — C withdraws ₹29,500 at an ATM

05:10 — A sends ₹50,000 on to F at Bank Z

desk closed until 10:00

10:20 — Desk opens at 10:00; first hold at 10:20. ₹500 is left

By the time anyone acts, **₹500** of the victim's money is left at Bank X. To be safe, the desk freezes Sri Murugan Fuels' whole account: **₹8,21,900 of his own money** locked over a ₹1,000 diesel sale, of which ₹500 was the victim's.



One complaint, nine parties, seven phases

Report



Victim calls 1930; the police operator raises a ticket on the national portal

1 Chasing the money



Bank nodal officer finds the payment, holds what is left, reports onward hops and exits

BUILD ZONE

2 Registering the case



Police register the FIR; e-Zero FIR above a loss threshold

3 Contesting the hold



Frozen account holders contest; grievance officer and IO verify in 15 days

IMPACT ZONE

4 Returning the money



IO works out each victim's share; bank pays on order

5 Going to court



Magistrate or Lok Adalat orders release

6 Closing the case



Unclaimed money; oversight reports

Inside the Layer-1 bank, one ticket passes through three teams, by hand

Portal desk
reads the ticket and notice

›

Core banking team
finds the UTR, marks the hold

›

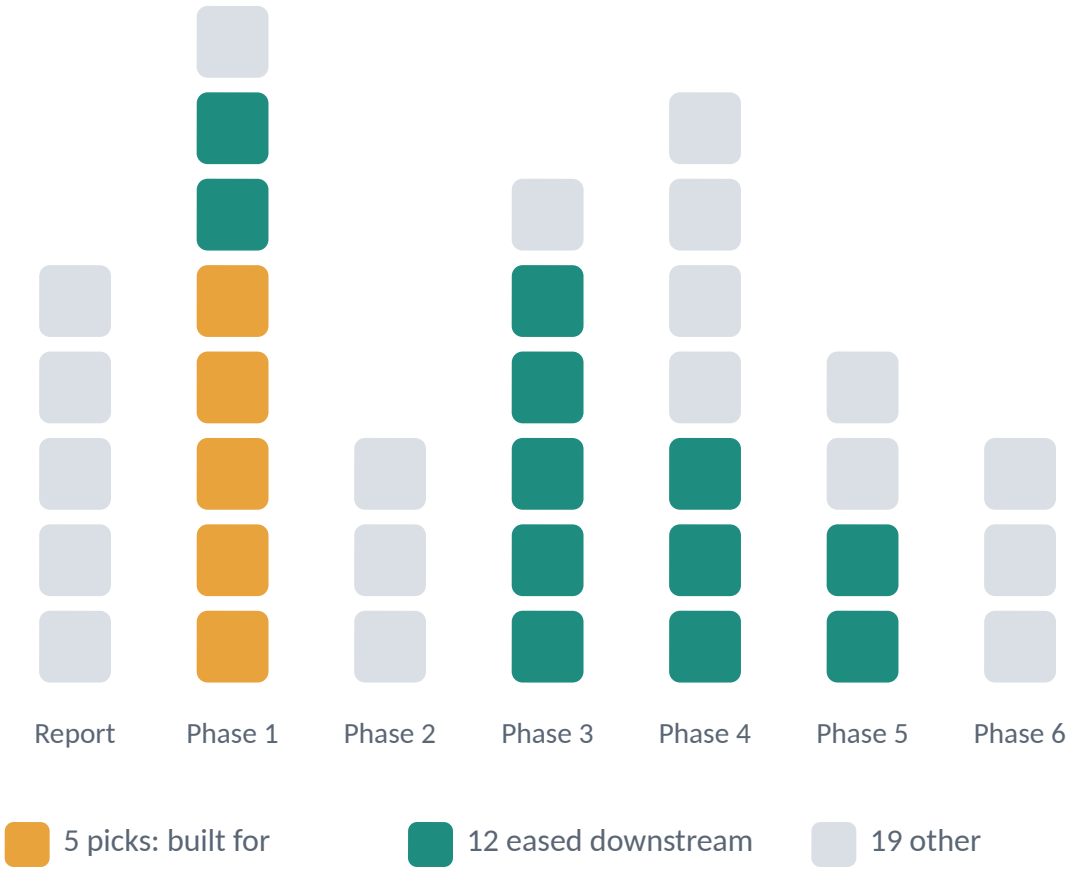
Portal desk
types every onward transfer and exit

›

Branch
learns by screenshot

The SOP says "real-time" with no minutes set; the desk often keeps bank hours. Full map, about 30 steps: docs/PROBLEM_DISCOVERY.md

36 pain points; we build for the five in Phase 1



The five picks, all at the bank nodal officer

1. Tickets arrive at night; desks keep bank hours
2. Each UTR is looked up and held by hand
3. Every onward transfer, exit and KYC packet is typed by hand
4. No rule for how much of mixed money is the victim's
5. Honest receivers deep in the trail are frozen

1 to 3 are speed. 4 and 5 are precision. Phase 1 is not solved, data-heavy, and has one clear role to take over.

What we built

An agent that works the nodal officer's tickets the moment they land, at any hour. It traces the victim's money, places the holds the rules already require, and hands the officer a ready proposal, with reasons, for every judgment call. Each hold is sized to the victim's share.

Automatic, at any hour

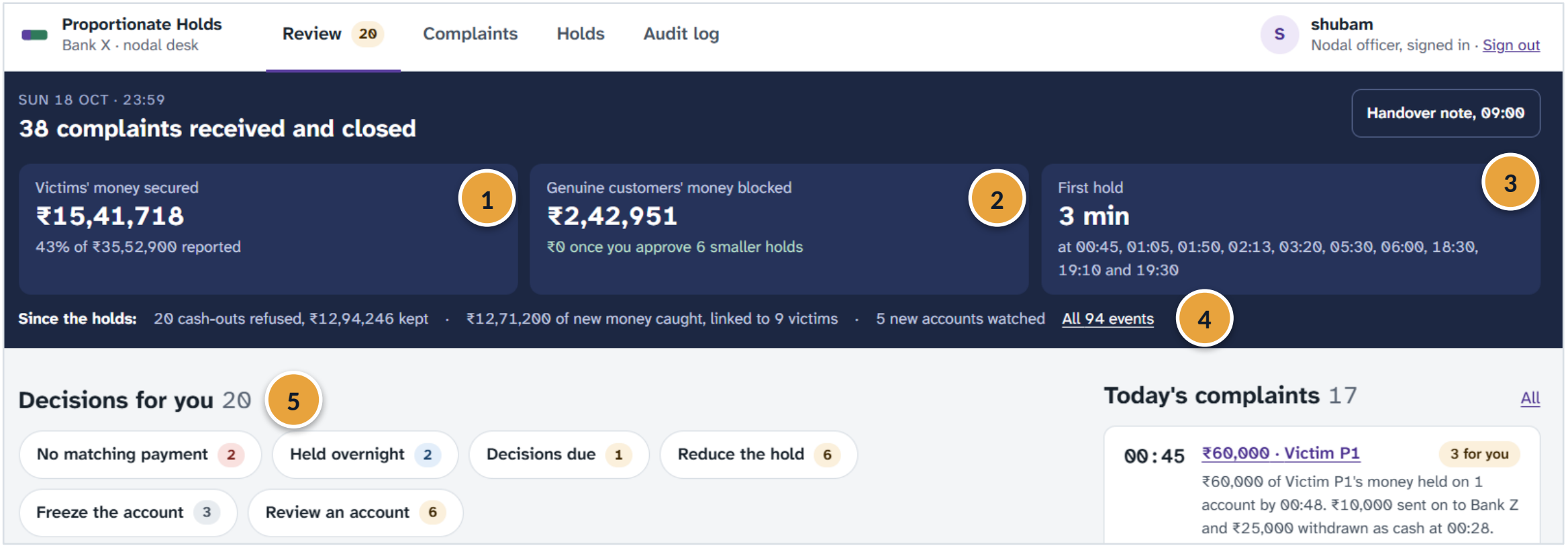
- Places the mandatory holds, amount only
- Records ATM, cash and crypto exits
- Files onward entries for the next bank
- Files KYC packets for the police
- Sends back tickets that match no payment
- Splits held money between victims
- Watches touched accounts, logs refused attempts

The officer decides

- Reduce a hold to the victim's share
- Full freeze: proposed, never placed by the system
- Larger hold, when fraud money leaves fast
- Unmatched notice: take the near match or leave it
- Release after 75 days with no police direction

Maker, checker, approver: the agent only proposes. A deterministic checker with its own database user validates every proposal and acts. The officer signs the five judgment calls.

23:59: 38 complaints worked, only the judgment calls wait



1 **Victims' money secured**
₹15,41,718 held across the day, at any hour

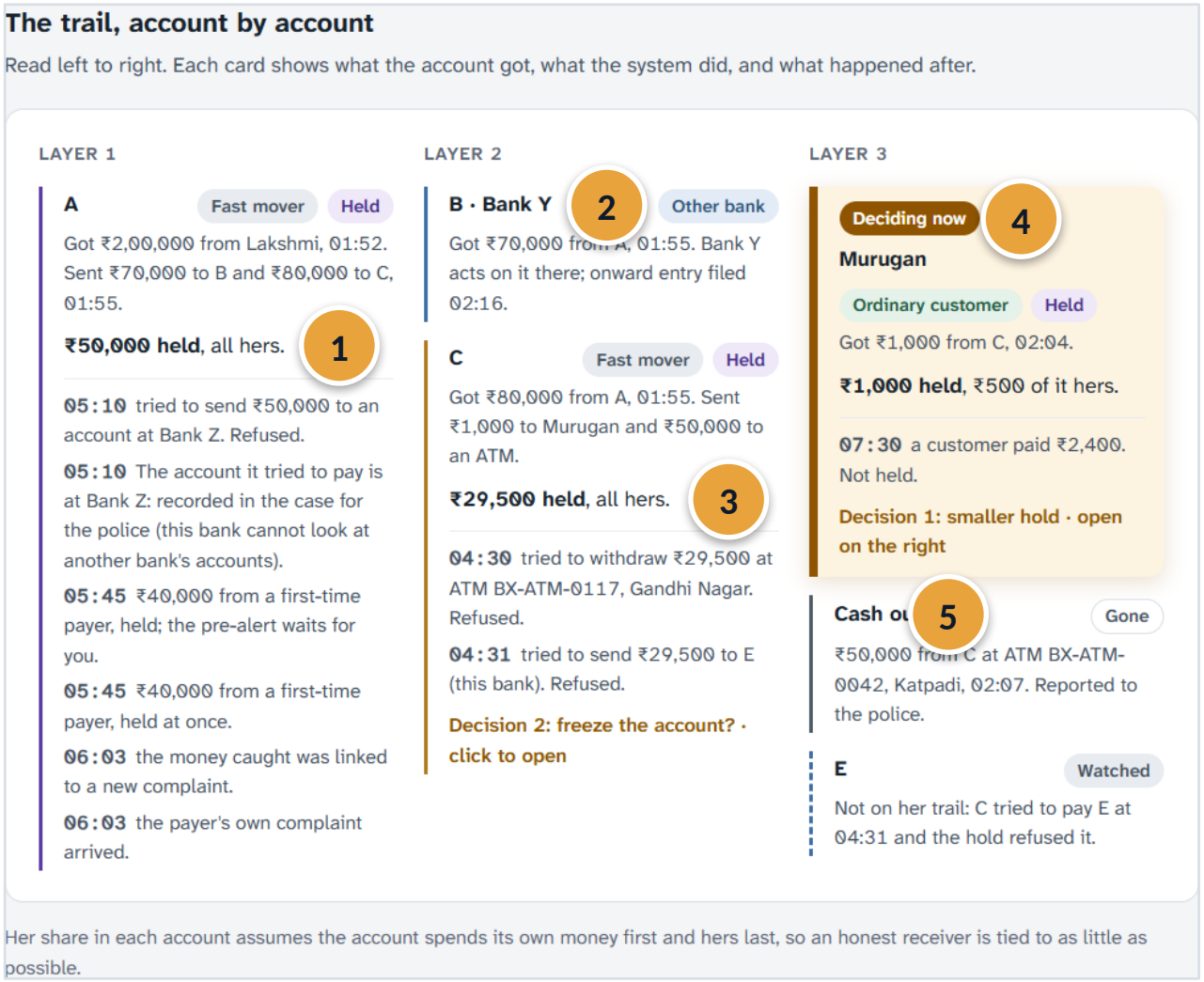
2 **Genuine customers' money**
Honest customers' money in the holds, shown next to the victims'

3 **First hold**
3 minutes after each ticket, from 00:45 to 19:30

4 **Since the holds**
20 cash-outs refused; ₹12,94,246 kept in place

5 **Decisions for you**
Only judgment calls wait: reduce, freeze, unmatched notice

02:13: the whole trail traced and held by 02:16



1 **Layer 1: A**
₹50,000 held at 02:16, all of it Lakshmi's; A's 05:10 transfer refused

2 **Other bank**
B is at Bank Y: the onward entry is filed at 02:16 and Bank Y acts on it

3 **C, Fast mover**
₹29,500 held; the 04:30 ATM withdrawal and 04:31 transfer refused

4 **Deciding now**
Murugan: ₹1,000 held, ₹500 of it hers; the smaller hold waits for the officer

5 **Cash out**
₹50,000 left at an ATM at 02:07, before the ticket; recorded for the police

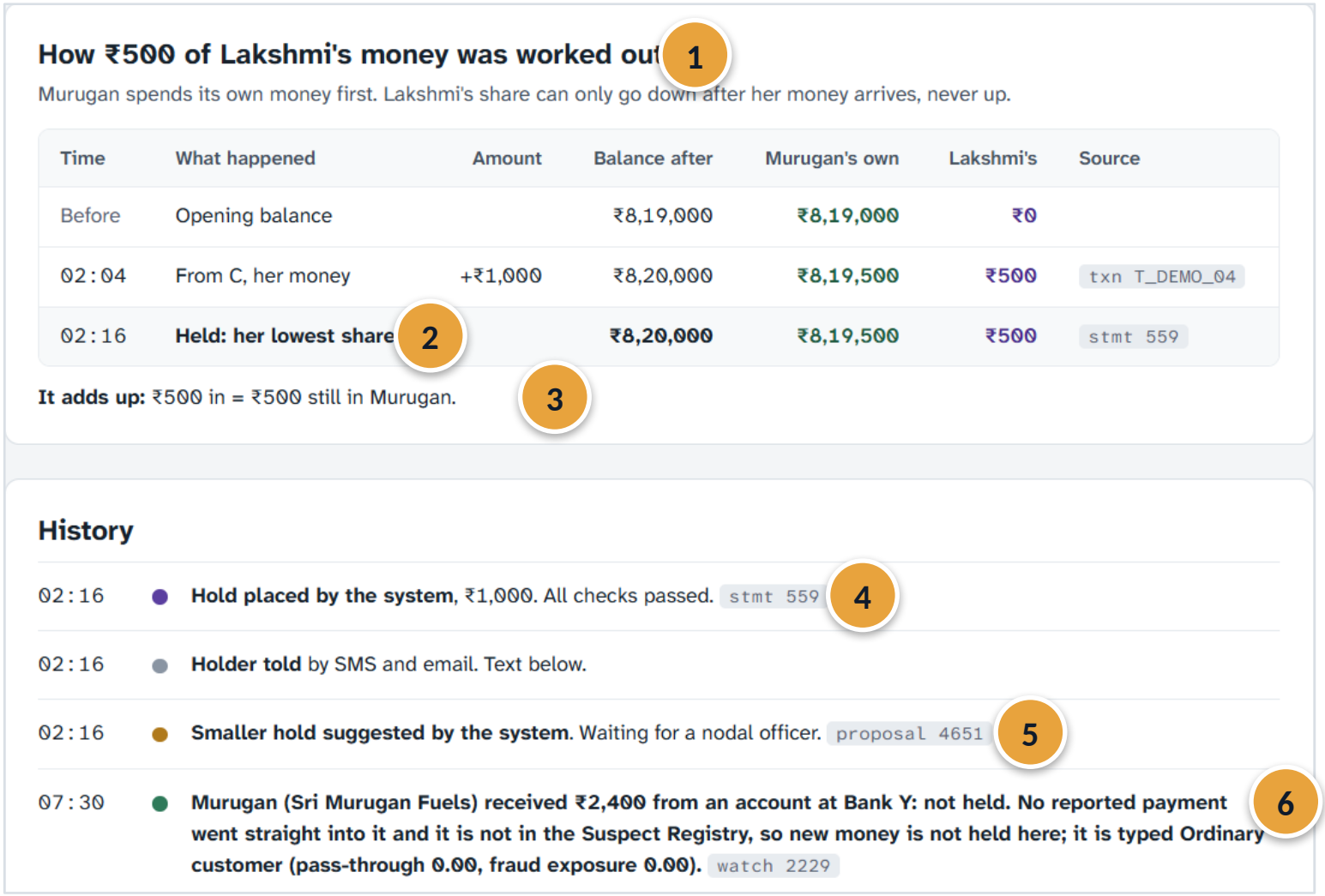
Every step the agent took, tied to an Exasol statement

What the system did — 11 steps in 11.6 s · fixed order · checker passed 8 of 8

1	**Read the ticket** police ticket from 1930: UTR IMPS2026101801520001, ₹2,00,000 into ACC_A, notice VLR/CCPS/2026/1018-01		
2	**Matched the payment** exact match: transaction T_DEMO_01	`stmt 713`	
3	**Looked for earlier complaints** a new transaction for this bank	`stmt 713`	
4	**Traced the money** 3 hops into this bank's accounts, 1 to other banks, 1 exits; 3 hops deep	`stmt 731`	
5	**Worked out her share** victim's money by account (LIBR): B ₹70,000, A ₹50,000, EXIT:ATM:BX-ATM-0042 ₹50,000, C ₹29,500, Murugan (Sri Murugan Fuels) ₹500	`stmt 755`	
6	**Read 90 days of history** account evidence at this bank: A: Fast mover (pass-through 0.85, fraud exposure 0.01); C: Fast mover (pass-through 0.86, fraud exposure 0.13); Murugan (Sri Murugan Fuels): Ordinary customer (pass-through 0.00, fraud exposure 0.00)	`stmt 787`	
7	**Checked the rules** the rules allow 5 automatic actions and 1 decisions for the officer	`stmt 807`	

1. **What the system did**
   11 steps in 11.6 s, in a fixed order; the checker passed 8 of 8
2. **Matched the payment**
   The notice is matched to the exact transaction before anything is held
3. **Traced the money**
   3 hops in this bank, 1 to another bank, 1 exit, in one statement
4. **Worked out her share**
   Her share per account (LIBR), in a Python UDF inside Exasol
5. **Read 90 days of history**
   Pass-through and fraud exposure for each account, in SQL
6. **Checked the rules**
   5 automatic actions, 1 decision for the officer

Murugan keeps his account: hold only the victim's ₹500



How ₹500 of Lakshmi's money was worked out

Murugan spends its own money first. Lakshmi's share can only go down after her money arrives, never up.

Time	What happened	Amount	Balance after	Murugan's own	Lakshmi's	Source
Before	Opening balance		₹8,19,000	₹8,19,000	₹0	
02:04	From C, her money	+₹1,000	₹8,20,000	₹8,19,500	₹500	txn T_DEMO_04
02:16	**Held: her lowest share**		**₹8,20,000**	₹8,19,500	₹500	stmt 559

It adds up: ₹500 in = ₹500 still in Murugan.

History

02:16 ● **Hold placed by the system**, ₹1,000. All checks passed. stmt 559

02:16 ● **Holder told** by SMS and email. Text below.

02:16 ● **Smaller hold suggested by the system.** Waiting for a nodal officer. proposal 4651

07:30 ● **Murugan (Sri Murugan Fuels) received ₹2,400 from an account at Bank Y: not held. No reported payment went straight into it and it is not in the Suspect Registry, so new money is not held here; it is typed Ordinary customer (pass-through 0.00, fraud exposure 0.00).** watch 2229

1. **How ₹500 was worked out**
   His own money leaves first, so only ₹500 of the ₹1,000 is Lakshmi's
2. **Held: her lowest share**
   Row by row, each step linked to its transaction
3. **It adds up**
   ₹500 in = ₹500 still in Murugan, to the paisa
4. **Hold placed by the system**
   ₹1,000, amount only, never his account; stmt 559
5. **Smaller hold suggested**
   Reducing to ₹500 waits for the officer, who signs it by name
6. **07:30, not held**
   A real customer's ₹2,400 goes through, and the log says why

C tried the ATM at 04:30: refused, and a freeze proposed

How ₹29,500 of Lakshmi's money was worked out

C spends its own money first. Lakshmi's share can only go down after her money arrives, never up.

Time	What happened	Amount	Balance after	C's own	Lakshmi's	Source
Before	Opening balance		₹500	₹500	₹0	
01:55	From A, her money	+₹80,000	₹80,500	₹500	₹80,000	txn T_DEMO_03
02:04	To Murugan. C's own ₹500 leaves first, then ₹500 of hers	−₹1,000	₹79,500	₹0	₹79,500	txn T_DEMO_04
02:07	Cash at ATM BX-ATM-0042	−₹50,000	₹29,500	₹0	₹29,500	txn T_DEMO_05
02:16	**Held: her lowest share**		**₹29,500**	₹0	₹29,500	stmt 537

It adds up: ₹80,000 in = ₹29,500 still in C + ₹500 sent to Murugan + ₹50,000 taken as cash.

Freeze — Waiting

Freeze C's whole account?

C ACC_C · savings, opened 2026-10-04

Freeze C | Keep the hold only

	Now	With a freeze
Money in C	₹29,500 held	₹29,500 held
New money in	Can leave	**Blocked**

1 **It adds up**
₹80,000 in = ₹29,500 held + ₹500 to Murugan + ₹50,000 cash

2 **Own money first**
C's own ₹500 left first, so all ₹29,500 left in C is hers

3 **Freeze C's whole account?**
Proposed after the refused ATM try; never placed by the system

4 **Freeze C or keep the hold**
The officer decides, with the reasons and the case against

5 **With a freeze**
The only change: new money into C is blocked too

Ready for the RBI draft: holds before a complaint

On an account a complaint names directly, or one in the Suspect Registry, that scores as Pipe for fraud or Fast mover, a first-time payment of ₹1,000 or more is held the minute it lands, and someone who can reach the payer is told.

rbi_draft RBI draft KYC Amendment Directions, 11 Sep 2026

- The payment's own amount is held, up to the free balance
- The holder is told; 20, 30 and 60-day clocks run
- A kept hold is referred to the police through the portal

current_sop Today's SOP

- New money is held only within an existing police notice's hold
- No new legal basis needed
- One setting switches between the two modes

Never on deeper layers, on accounts flagged only by our own score, or on nodal, pool and escrow accounts. **The draft directions are not final law,** so the bank can run on today's SOP until they are.

On the demo day: 12 holds placed before anyone complained; 9 of the 10 closed were confirmed by a victim's complaint, a median 2 h 2 min later. *Synthetic data: what a bank would track, not results.*

How it works: the numbers are computed inside Exasol

Simulated national portal
PH_PORTAL

Tickets in, onward entries and KYC packets out. The only link between banks

Exasol Personal · one schema per bank: PH_X, PH_Y, PH_Z

Intake	matches the notice to a transaction
Trace	link table, then CONNECT BY NOCYCLE, loop-safe
Attribute	Python UDF replays balances in integer paise (LIBR)
Score	two scores from SQL signals over 90 days of history
Rules	automatic, officer, or forbidden
Checker	eight checks incl. conservation to the paisa; no change if any fails

Maker, checker, officer
Agent proposes; checker acts with its own database user; officer signs the judgment calls

Hold ledger and audit log
Insert-only; every number links to its Exasol statement ID

1,000
tickets at once, all worked within the hour

4.49M
AMLworld transactions as background traffic

5.76 s
median work per ticket, intake to holds

0
checker failures, 0 errors

Where AI and ML fit, and where they don't

Local model, optional

- Runs on the laptop with Ollama: free, offline, no data leaves the bank
- Picks the next tool from those the rules allow
- Words the reduce and freeze reasons and the morning log
- Never calculates or decides. A figure it wasn't given means its text is thrown away
- Switched off, a fixed pipeline places the same holds

Learned scoring, gated

- Gradient-boosted trees trained on AMLworld rank held-out accounts at AUC 0.870, against 0.537 for the hand-set weights
- Live holds use explainable weights: every hold is defended signal by signal, as courts ask
- Retraining runs inside Exasol (make learn); a new version needs the tests, no accuracy loss and a named approval

Not a mule detector. MuleHunter.AI and the I4C Suspect Registry already do that. The question here is how much to hold, and whether the bank can defend it.

Manual desk vs the system, on the same day

	Manual desk	The system
LAKSHMI'S 02:13 COMPLAINT		
Her money held	₹500 Bank X	₹1,20,000 60%, both banks
Murugan's own money blocked	₹8,21,900 whole account	₹500 ₹0 once approved
First hold	10:20 8 h 7 min	02:16 3 min
ALL 38 COMPLAINTS AT BANK X		
Victims' money held	24% ₹8,44,772	43% ₹15,41,718
First hold, after-hours complaints	4 h 20 to 15 h 50	3 min

Limits: synthetic data; the portal, core banking and other banks are simulated; a real bank sees only its own accounts; cash already withdrawn is gone; the manual desk is a stated model (open 10:00 to 18:00, about 20 min for the first account and 15 min for each next, whole-account freezes), not a measurement.

Held in the golden hour. Only the victim's share.

Code github.com/yvssshubam/exathon-_2026

Demo video linked in the README

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